Sales Optimization Overview

Sales & Profitability Optimization

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Key Business Challenges Overview

- Revenue is strong but profit is inconsistent.
- Certain regions and categories underperform.
- Aggressive discounting may be reducing margins.
- Business needs data-driven pricing optimization.

Understanding Revenue vs. Profit Issues

Revenue is strong; profit remains inconsistent.

Dataset Overview: Sales Transactions Analysis

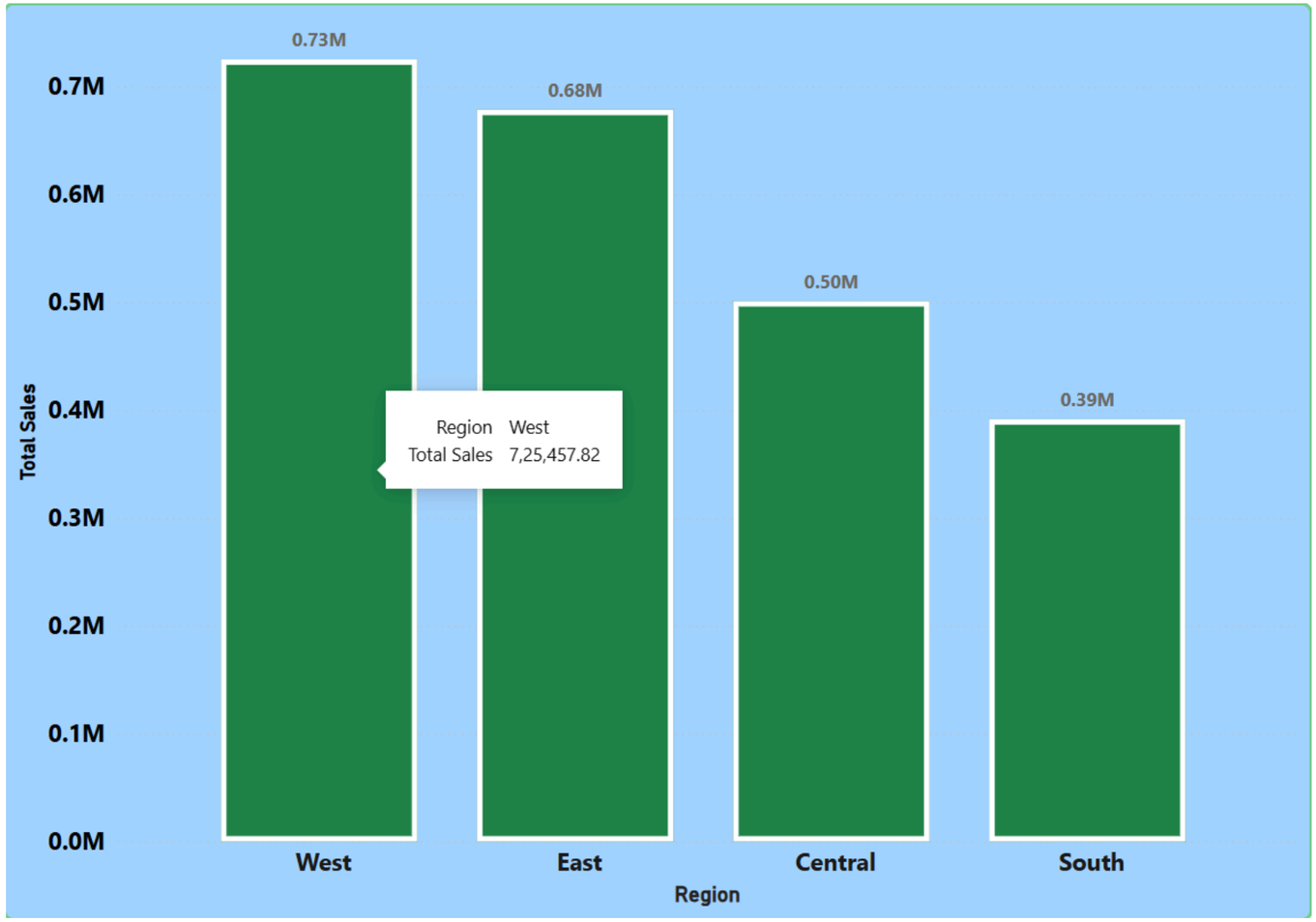
- Key features: Sales, Profit, Discount, Category, Region, Customer Segment

Comprehensive Analysis of Sales Data

9,994 sales transactions provide valuable insights.

Regional Performance

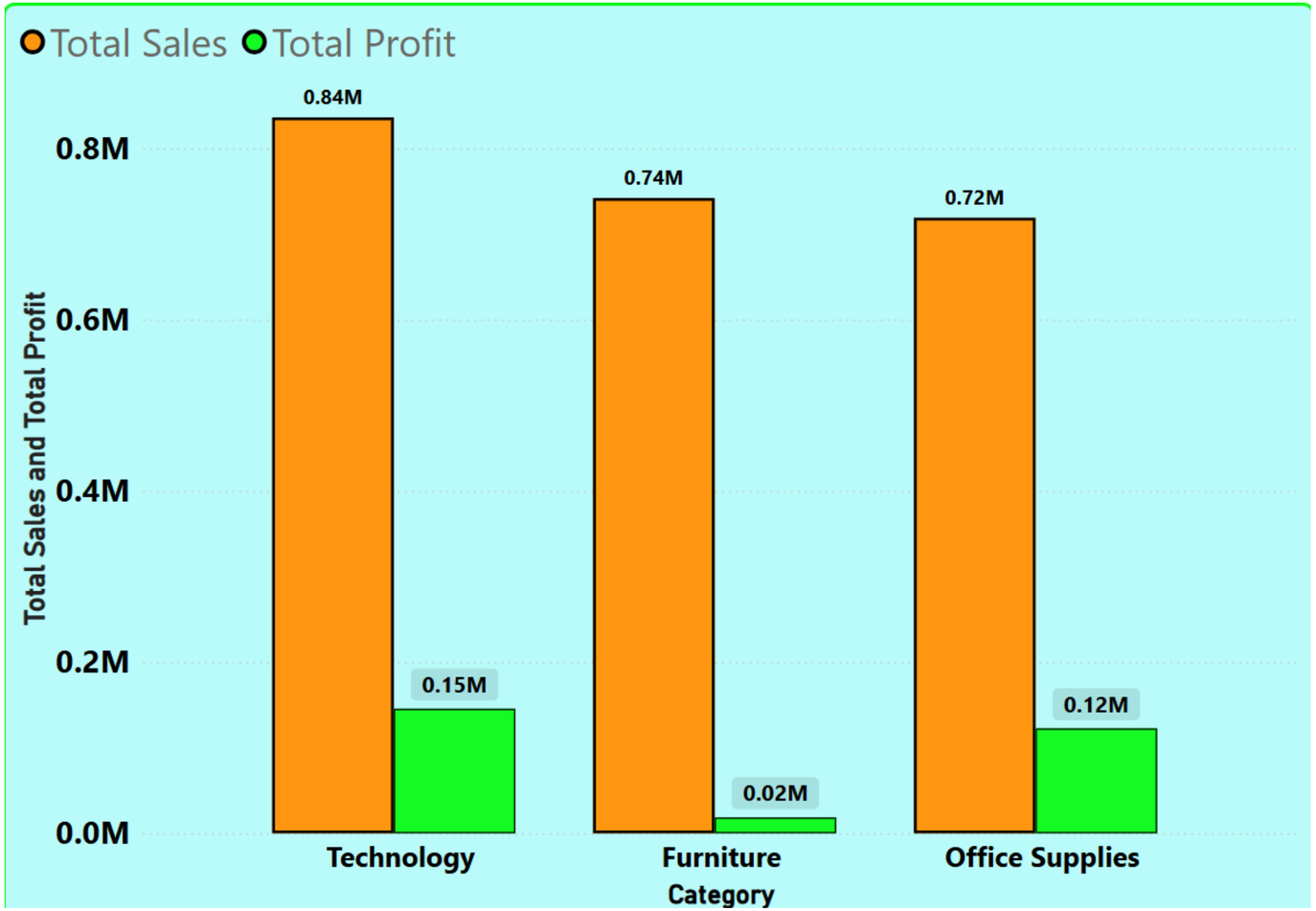
The West region has shown **remarkable profitability**, outperforming others significantly, highlighting the need for strategic focus and resource allocation.



Category Performance

The **Technology category** consistently outperforms others, driving significant profit margins and necessitating focused strategy adjustments going forward.

- **Furniture** generates lower margins.
- Category-level strategy required.



Discount Impact

Statistical Validation of Pricing Effects

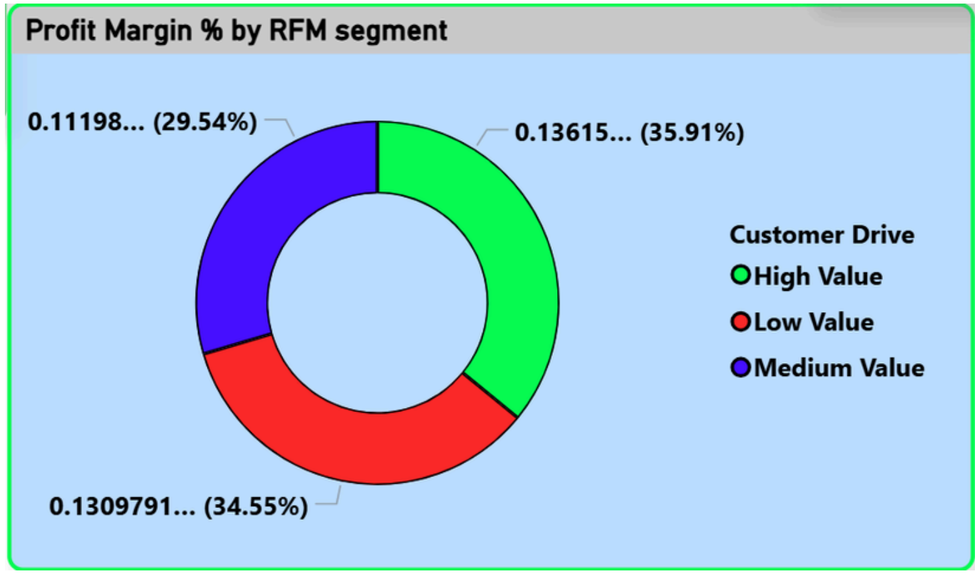
High discounts have a **significant negative** impact on average profit, as supported by statistical analysis indicating strong evidence against discounting strategies that harm profit margins.

```
t_stat, p_value = stats.ttest_ind(high_profit, low_profit)

print("T-Statistic:", t_stat)
print("P-Value:", p_value)
```

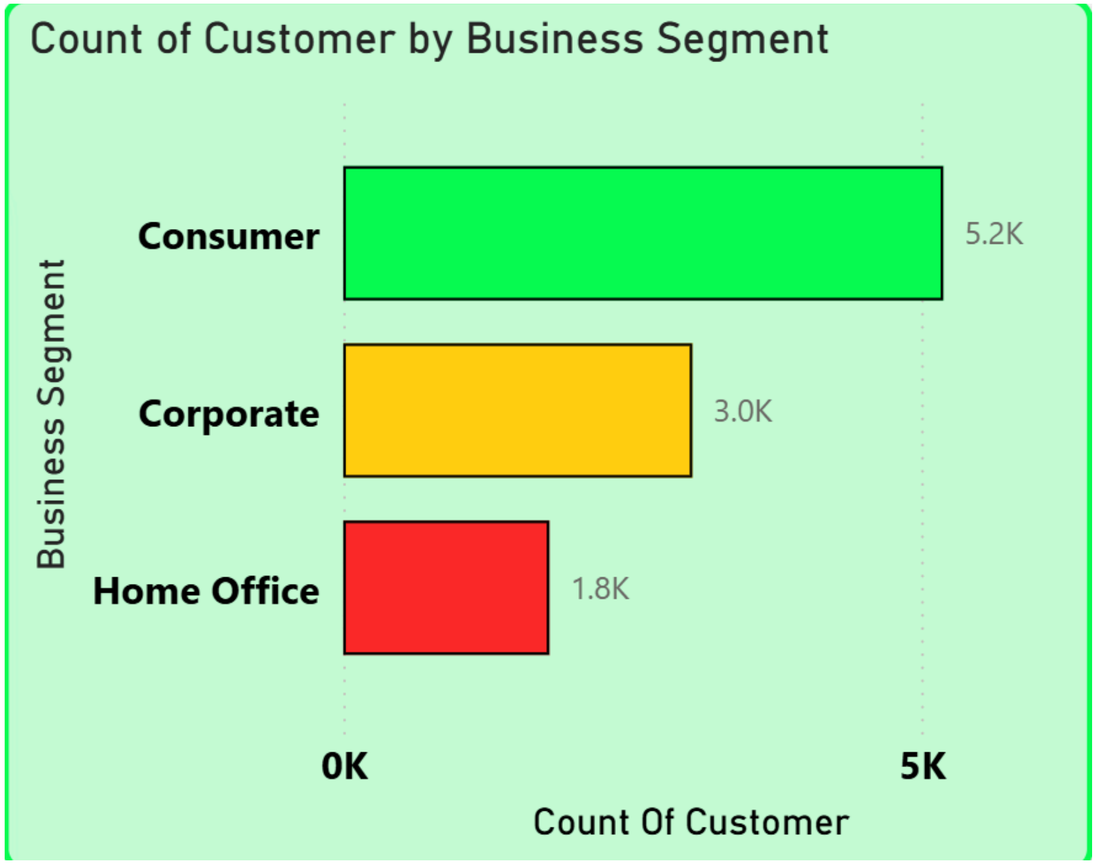
```
T-Statistic: -22.134190085874568
P-Value: 5.092886028896182e-106
```

Customer Insights and Segmentation



Understanding High-Value Customers

High-value customers are essential for business growth.



Understanding Business Segment Customers

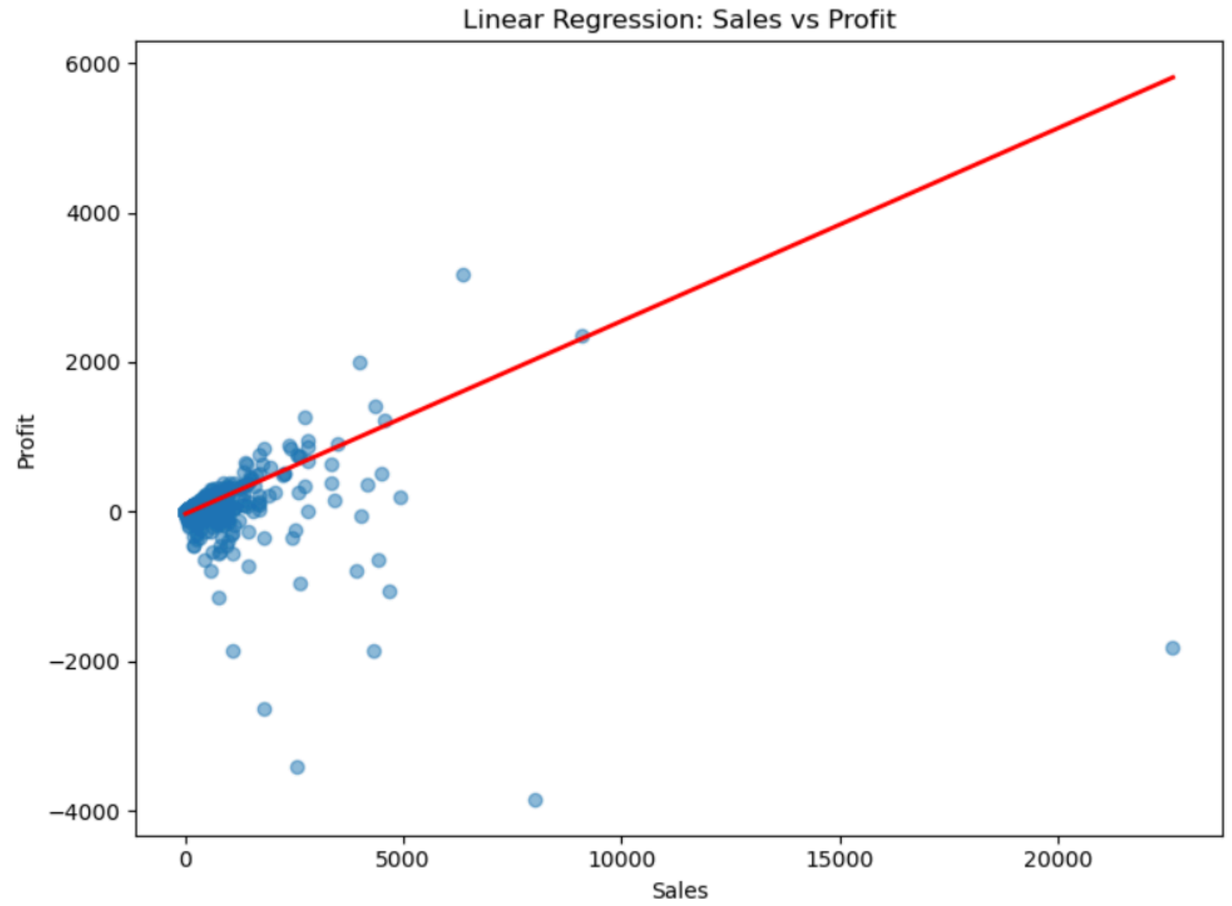
Some segments show lower engagement

Predictive Modeling Insights for Profitability

Understanding Non-Linear Profit Influences

Evaluating complex factors affecting profit margins.

Profitability influenced by non-linear factors.



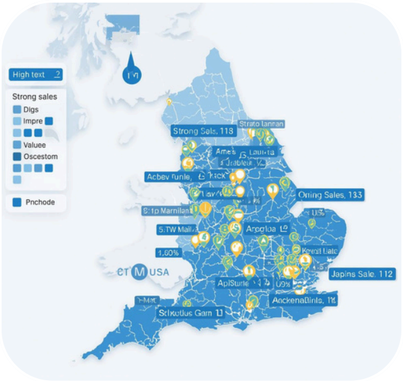

```
Improved R2: 0.7382052067825515
Improved MAE: 17.301706442841486
```

Optimize Discounts

Focus on effective pricing to enhance profits.

High-Margin Focus

Prioritize categories with the best profitability margins.



Strengthen Regions

Reinforce strategies in areas generating the most profit.

Target Customers

Engage and retain high-value customers effectively.

Monitor Segments

Keep track of underperforming business areas closely.

Call to Action

Drive sustainable profitability growth

Adopt a data-driven pricing and customer strategy to improve sustainable profitability